PROJECT FINAL REPORT

Auto Ticket Classification using Flow Designer | Comprehensive Documentation Template

Project Title:	Auto Ticket Classification using Flow Designer
Domain / Platform:	ServiceNow ITSM & Process Automation (Flow Designer)
Team Id:	SWTID-2026-6328
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1. INTRODUCTION

1.1 Project Overview

> **[INSTRUCTION / GUIDELINE]** *Describe the general background of the project, institution/organization context, and core operational challenge.*

The school IT helpdesk receives multiple incident requests daily from students and teachers, such as Wi-Fi issues, projector failures, password problems, and slow computers. Currently, IT staff manually reviews each request and assigns a category and subcategory, which is time-consuming, error-prone, and inefficient.

This project automates ticket classification using ServiceNow Flow Designer by analyzing keywords in the incident's short description and description fields, assigning the correct category and subcategory automatically upon ticket submission.

1.2 Purpose

> **[INSTRUCTION / GUIDELINE]** *Outline the core objectives and intended outcomes of the project.*

The primary purpose of this project is to implement a robust, no-code automated ticketing solution within ServiceNow. Key objectives include:

- Automatically classify incidents at the time of creation based on short description keywords.
- Reduce manual effort and operational overhead for IT helpdesk agents.
- Improve ticket routing efficiency, response times, and overall ticket tracking.
- Provide an easily maintainable, scalable, and no-code workflow standard for educational institutions.

2. IDEATION PHASE

2.1 Problem Statement

School IT helpdesks face high ticket volumes daily regarding network access, hardware faults, login credentials, and system latency. Manual intake leads to bottlenecking, delayed ticket assignment, inconsistent categorization, and high agent fatigue. An automated keyword-driven classification mechanism is required to streamline the intake process.

2.2 Empathy Map Canvas

> **[INSTRUCTION / GUIDELINE]** *Detail user perspectives (Students, Teachers, IT Helpdesk Agents) across Says, Thinks, Does, and Feels.*

Quadrant	User Role (Student / Teacher)	IT Helpdesk Staff
Says	"My Wi-Fi isn't connecting", "Projector is off", "I forgot my password."	"We spend too much time manually reading and tagging tickets before solving them."
Thinks	"Why does it take so long to get help for simple password resets?"	"We need a standardized, auto-categorized inbox to meet SLAs."
Does	Submits short incident descriptions with basic keywords.	Manually opens tickets, selects Category and Subcategory, assigns groups.
Feels	Frustrated by downtime and administrative delays.	Overwhelmed by repetitive administrative tagging and backlog volume.

2.3 Brainstorming

During the ideation phase, multiple approaches were evaluated for ticket classification:

1. Manual Assignment (Baseline): High error rate, slow SLAs.
2. Scripted Business Rules (JavaScript): Effective, but requires code maintenance.
3. Flow Designer (No-Code Workflow): Best balance of speed, maintainability, visual debugging, and zero script requirement.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. Ticket Submission: Caller logs into ServiceNow portal and enters Caller Name and Short Description.
2. Keyword Parsing: System triggers Flow Designer flow on ticket creation.
3. Dynamic Classification: Flow checks keywords (e.g., 'WiFi', 'Projector', 'Password', 'Slow') and populates Category and Subcategory.
4. Automated Notification: System sends confirmation email to the caller.
5. Resolution: IT Agent receives pre-classified ticket for immediate action.

3.2 Solution Requirement

The solution must meet the following technical and operational requirements:

- Automatically classify IT tickets based on keywords in Short Description.
- Assign both Category and Dependent Subcategory without manual intervention.
- Enforce dependent choice list logic in the UI (e.g., Network -> Wi-Fi).
- Send an automated email notification to the caller upon creation.
- Store ticket data in a dedicated custom table (`u_incident_workflow`) with auto-numbering (prefix 'INC').
- Package all configurations into an Update Set for environment portability.

3.3 Data Flow Diagram

> **[INSTRUCTION / GUIDELINE]** *Illustrate or describe the data inputs, flow logic, decision nodes, and database output.*

[Caller Input: Short Description] ---> (Trigger: Record Created where Category is Empty) ---> [Flow Designer Condition Logic] ---> [Update Record: Category & Subcategory] ---> [Action: Send Email Notification] ---> [Database: u_incident_workflow Record Updated]

3.4 Technology Stack

Component / Layer	Technology / Specification
Platform	ServiceNow Enterprise Cloud Platform
Core Application	IT Service Management (ITSM) / Custom Table Engine
Workflow Engine	ServiceNow Flow Designer / Workflow Studio
Deployment & Migration	ServiceNow Local Update Sets (XML Export/Import)

4. PROJECT DESIGN

4.1 Problem Solution Fit

By matching specific high-frequency user keywords directly to standard ITSM categories, 90%+ of initial intake tickets are categorized instantaneously, eliminating manual triage latency.

4.2 Proposed Solution & Data Schema

The implementation consists of a custom table with structured dictionary entries, choice lists, and field dependencies:

Field Label	Data Type	Reference / Choice Options
Number	Auto Number	Prefix: INC, Digits: 5, Start: 500
Caller	Reference	sys_user table
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice (Dependent)	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String	User-entered summary
Description	String	Detailed issue information
State	Choice	New, In progress, On hold, Resolved, Closed
Assigned Group	Reference	sys_user_group table
Assigned to	Reference	sys_user table

4.3 Solution Architecture & Flow Logic

The Flow Designer logic is structured as follows:

• Trigger: Record Created on `Incident WorkFlow [u_incident_workflow]` where Category is empty.
• Decision Node 1 (Wi-Fi): IF Short Description contains 'Wi-Fi' OR 'Network' -> Set Category = Network, Subcategory = Wi-Fi.
• Decision Node 2 (Projector): ELSE IF Short Description contains 'Projector' -> Set Category = Hardware, Subcategory = Projector.
• Decision Node 3 (Password): ELSE IF Short Description contains 'Forgot password' -> Set Category = Access, Subcategory = Forgot Password.
• Decision Node 4 (Performance): ELSE IF Short Description contains 'Slow computer' -> Set Category = Performance, Subcategory = Slow Computer.
• Action: Send Email to Caller (To: Trigger Record -> Caller -> Email, Subject: 'Your Request for the issue has been submitted').

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning & Phased Execution

The project was executed across five sequential phases:

Phase	Phase Name	Key Deliverables & Activities
Phase 1	Requirement Analysis & Planning	Define use case, business requirements, and create 'Project Update Set'.
Phase 2	Backend Development & Configuration	Create custom table (`u_incident_workflow`), configure fields, auto-numbering, and choice dependencies.
Phase 3	Automation using Flow Designer	Build flow 'Auto Classify School IT Tickets', configure trigger, keyword logic branches, and email action.
Phase 4	Testing, Validation & Security	Execute end-to-end test scenarios (Wi-Fi, Projector), verify email logs, validate ACLs.
Phase 5	Deployment & Final Documentation	Mark Update Set as 'Complete', export XML, and compile project report.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing & Data Validation

Validation checks performed during execution:

- Auto-Number Generation: Verified unique incrementing IDs starting at INC00500.
- Dependent Choice Integrity: Confirmed Subcategory dropdown options dynamically restrict based on selected Category.
- Flow Trigger Execution: Verified flow executes within < 1 second of record insert.
- Email Delivery Log: Verified generated notification records in System Logs -> Emails.

7. RESULTS

7.1 Output Screenshots & Test Cases

> **[INSTRUCTION / GUIDELINE]** *Insert screenshots of ServiceNow forms, Flow Designer canvas, email logs, and XML export files here.*

Test ID	Input Short Description	Expected Category	Expected Subcategory	Pass / Fail
TS-01	WiFi not working in library	Network	Wi-Fi	PASS
TS-02	Projector not turning on	Hardware	Projector	PASS

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- Zero-Code Maintenance: Easily modified by non-developer ServiceNow administrators.
- Instant Triage: Cuts intake categorization time from minutes to zero delay.
- Data Standardisation: Eliminates manual user misclassifications.
- Portability: Complete configuration encapsulated in a single Update Set XML file.

8.2 Disadvantages & Limitations

- Keyword Sensitivity: Relies on exact keyword matches in short description.
- Unlisted Terms: Complex or misspelled descriptions fall back to unclassified or require default handling.

9. CONCLUSION

The Auto Ticket Classification project successfully delivers a robust, automated ITSM solution on the ServiceNow platform. By combining custom table architecture, choice field dependencies, and Flow Designer keyword automation, the manual operational burden on school IT helpdesk staff is substantially eliminated while ensuring high data fidelity.

10. FUTURE SCOPE

- Predictive Intelligence / Machine Learning: Transitioning from keyword logic to ServiceNow Predictive Intelligence NLU models.
- Automated Assignment Groups: Routing tickets directly to tier-1 / tier-2 support groups based on category.
- SLA Tracking: Attaching Service Level Agreements (SLAs) automatically based on assigned issue priority.

11. APPENDIX

11.1 Update Set Export File

Filename: sys_remote_update_set_a7b40de6fc590ceaad36c.xml
State: Complete
Application: Global